# BIOMETRIC BLOOD GROUP IDENTIFICATION SYSTEM

Final Year B.E. Major Project • Deep Learning AI Analysis Report

**MEDICAL DISCLAIMER:** AI-based preliminary estimation only. This result must be confirmed using a standard laboratory blood typing test before medical use. This system does not replace standard serological blood typing.

## 1. EXAMINATION & SAMPLE AUDIT DETAILS

| | | | |
|---|---|---|---|
| **Report ID:** | BG-000458 | **Timestamp:** | 2026-09-06 19:38:35 |
| **Input Source:** | MFS100 (Mock Mode) | **Evaluation Status:** | **NORMAL** |
| **Model Name:** | Champion Ensemble V2 (ResNet50 58% + VGG16 38% + EffNet 4%) | **Confidence:** | **66.98%** |

## 2. BIOMETRIC IMPRESSION & PREDICTED BLOOD GROUP

# Predicted Blood Group:

# O+

Confidence Score: **66.98%**

Threshold: 60.0%

Classification: High Confidence

Derived ABO Group: **O** (93.5% prob) | Rh Factor: **+**

## 3. 8-CLASS BLOOD GROUP PROBABILITY DISTRIBUTION

| A+ | A- | AB+ | AB- | B+ | B- | O+ | O- |
|---|---|---|---|---|---|---|---|
| 1.1% | 3.4% | 0.1% | 1.9% | 0.0% | 0.0% | 67.0% | 26.5% |